

Novel Graphical Tool for the Design of the Heat Integrated Water Allocation Networks

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A new graphical tool has been developed for the HEN design of WAHEN. An ΔH -F diagram is proposed to deal with the heat exchange matches. Nonisothermal stream mixing and splitting can be achieved by the transformation of stream composite curve in the ΔH -F diagram. A matching composite curve is proposed for the design of HEN with parallel structure. The structure of HEN can be simplified by adjusting the shape of the matching composite curve. Both small scale and large scale examples are illustrated. Results show that the proposed method is as good as the previous methods for small scale problem, while it gets a better result for large scale problem. © 2015 American Institute of Chemical Engineers AIChE J, 62: 670–686, 2016

Keywords: heat integrated water allocation networks, matching composite curve, parallel structure

Introduction

Heat integrated water allocation networks (abbreviated as WAHEN) are common in process industries, where water serves as a carrier of both contaminants and energy. Generally, WAHEN can be decomposed into two subsystems: water allocation networks (abbreviated as WAN) and heat exchanger networks (abbreviated as HEN). The design methodology for both WAN^{1–3} and HEN⁴ are well developed. It seems one can directly obtain the WAHEN structure by using the matured WAN and HEN design approach. However, if we treat these subsystems separately, additional energy and water penalty may be incurred. Therefore, the holistic design and optimization of WAHEN have drawn more and more attention during the last decade.

The design methodology of WAHEN can be categorized into two classes: conceptual design and mathematical programming. For mathematical programming methods, various superstructure features have been proposed to combine the models of WAN and HEN. Bagajewicz et al.⁵ proposed state-space representation for model statement, Leewongtanawit and Kim⁶ combined the WAN superstructure,⁷ heat transshipment model,⁸ and HEN superstructure model⁹ to formulate a MINLP model for simultaneous optimization of water and energy. Dong et al.¹⁰ presented a modified state-space superstructure model that captured more structural possibilities, but it is computational demanding. Subsequently, Zhou et al.^{11,12} proposed a multiscale state-space superstructure for the synthesis of integrated interplant water allocation networks and heat exchanger networks (IWAHENS) considering both fixed

flow rate processes and fixed contaminant-load processes. In addition, Bogataj and Bagajewicz,¹³ Ahmetovic and Kravanja,¹⁴ and Liao et al.¹⁵ extended the stage-wise superstructure¹⁶ to find simple but efficient WAHEN structures. It should be noted that these stage-wise-based superstructure are well performed among the above reported literature. Later, Ahmetovic and Kravanja extended their previous superstructure¹⁴ to involve process-to-process streams¹⁷ and water treatment units.¹⁸ Ibric et al.¹⁹ proposed a superstructure for the synthesis of pinched and threshold heat-integrated water allocation networks, including gradual heating and cooling of freshwater. This model can be extended to include wastewater treatment and recycling.²⁰ Note that, among their work,^{17–20} the two-step solution strategy is adopted to solve the MINLP, since the result of first step can be used as an initialization point in the second step. Recently, Liu et al.²¹ proposed a two-step approach. At first, the WAN is obtained by maximizing stream mixing and then, the remaining HEN is designed by pinch-based method. Ghazouani et al.²² introduced a new approach for synthesis of heat integrated resource allocation network with a modified transshipment model, which account for nonisothermal mixing by building a predefined temperature scale.

The conceptual-based methods were initiated by the research group from Manchester.^{23–25} They developed a two dimensional grid diagram to guide the allocation of water streams, and a separate system to help design the HEN. Sahu and Bandyopadhyay²⁶ obtained the energy target of WAHEN by incorporating pinch insights into linear programming formulations. Manan et al.²⁷ presented a new technique through a combination of numerical and graphical tools. The new graphical tool, termed as heat surplus diagram, was developed to guide water and energy reduction simultaneously, and provide insights on the energy demand as well as on stream matching scenarios. Wan Alwi et al.²⁸ combined the water and heat pinch diagrams to guide the design toward the minimum

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utility targets. Luo et al.^{29,30} discussed the effect of nonisothermal mixing on the energy consumption. Feng et al.³¹ analyzed the energy aspect of several water networks and concluded that the number of temperature fluctuations should be minimized. Recently, Hou et al.³² used the concentration potential concept³³ to build temperature and concentration order composite curves, which is suitable for multicontaminant cases.

It should be noted that most of the conceptual methods are focused on designing energy efficient WAN. Since there lies great opportunity in stream splitting and mixing inside the HEN of WAHEN, the optimized HEN structure could be more simplified than the conventional HEN structure. Although the structure possibility is richer, only the separate system concept^{23,25} has been developed for the conceptual design of HEN. In addition, the HEN designed by the separate system approach²⁵ are series HEN. But series structure is not the only structure for HEN and it is not always the simplest one. In this research, a novel graphical tool, ΔH -F diagram, is proposed to deal with the HEN design of WAHEN. HEN of parallel structure are obtained and simplified in the guide of ΔH -F diagram.

In the ΔH -F diagram, a hot stream composite curve, a cold stream composite curve as well as a matching composite curve are introduced to decide the number and load distributions of heat transfer matches.

Problem Statement

The general formulation of the heat integrated water allocation networks problem can be stated as follow:

Given is a set of water-using operations which require water of a certain temperature and quality. The freshwater source is specified with given temperature and contaminant concentration, while the effluent is only specified to be below a maximum temperature value. The goal is to obtain a heat integrated water allocation network that minimizes the total cost. In this article, special concern is focused on the construction of parallel HEN structure and its simplification.

Several assumptions are made to simplify the problem. The maximum inlet and outlet concentrations of contaminant, the contaminant mass load, and the operating temperatures for water-using operations are specified. The flow rate does not change through operations and the system is assumed to be operating continuously. In addition, the heat capacity for all streams is assumed to be constant at 4.2 kJ/kg·K, since the fluctuation ranges of concentration and temperature are very small.

Although simultaneous optimization of heat-integrated water allocation networks may get better results than sequential methods, it suffers from the computational burden, especially for the large scale problems. So far, the problems with scale of 8 or 15 operations are only solved by sequential methods. Therefore, it is still necessary to develop sequential methods^{15,19,23,25} for large scale problems, and this is the choice of this article. WAHEN can be decomposed into two interconnected blocks: the WAN and the HEN. These two blocks can be designed successively: WAN design first then followed by HEN design. Since there are plenty of work on the WAN design of WAHEN, we mainly focus on the design of HEN block to explore further opportunities of network structure simplification. For WAN design, we adopt the method proposed by Savulescu et al.,^{23,25} which will be reviewed in the next section before the HEN design.

Table 1. Stream Data Extracted from the WAN

Stream	T_S (K)	T_T (K)	F (kg/s)	ΔH (kW)
FW1	293	T_{FW1}	F_{FW1}	—
FW2	293	T_{FW2}	F_{FW2}	—
FW3	293	T_{FW3}	F_{FW3}	—
WW1	T_{WW1}	303	F_{WW1}	—
WW2	T_{WW2}	303	F_{WW2}	—
WW3	T_{WW3}	303	F_{WW3}	—
WW4	T_{WW4}	303	F_{WW4}	—

Methodology

Synthesis of water allocation networks

Savulescu et al.^{23,25} presented a systematical conceptual method for the WAHEN design. In their method, the WAN is obtained by a two-dimensional grid diagram and the concept of water mains, in which energy transfer is maximized by non-isothermal mixing of reuse streams. Since this approach not only achieve freshwater minimization but also may reduce energy cost in the HEN design, we will adopt this approach to obtain water allocation networks. The design procedure involves the following four steps:

1. Setup the two dimensional diagram
2. Connect operations with water mains and water distribution inside mains
3. Merge operations crossing boundaries
4. Remove intermediate water mains through water and energy analysis.

Similar to the work of Savulescu et al.,^{23,25} we will choose simple case to illustrate the generation of parallel HEN structures. In this case, the reuse streams are only involved in direct heat transfer. Therefore, the indirect heat transfer (i.e., HEN) only includes freshwater streams as cold streams and wastewater streams as hot streams. For the proposed method, freshwater is assumed as cold streams since no water consuming operations that operate below 293 K is found in all the published HIWAN examples and several process industries such as the oil refinery, the pulp, and paper industry, and the olefin industry. In addition, the starting temperature of freshwater streams and the ending temperature of wastewater streams are specified at 293 K and 303 K, respectively. After the WAN is obtained, let's consider the design of HEN in parallel structure in the next section.

Presentation of the parallel HEN structure

Before presenting the parallel structure, let's extract the hot and cold stream data from the water allocation network and analyze the mass balance and energy balance of the HEN. From the mass balance, we can see that the flow rate of freshwater and wastewater are the same. From the overall energy balance, both heating and cooling utility consumptions can be calculated. To simplify the design procedure, hot utility is applied to the hottest ending among cold streams, while cold utility is supplied to the coldest starting between hot streams. After utility matching, the remaining hot and cold streams can completely cover their enthalpy duties by direct heat exchanges.

An illustrative case of these remaining hot and cold streams is presented in Table 1, where the utility and their corresponding matching parts are not shown. From Table 1, the illustrative HEN case includes three freshwater streams and four wastewater streams. The freshwater streams are denoted by FW1, FW2, and FW3, respectively, while wastewater streams are denoted by WW1, WW2, WW3, and WW4, respectively. Because utility matching parts are not shown, the total cooling

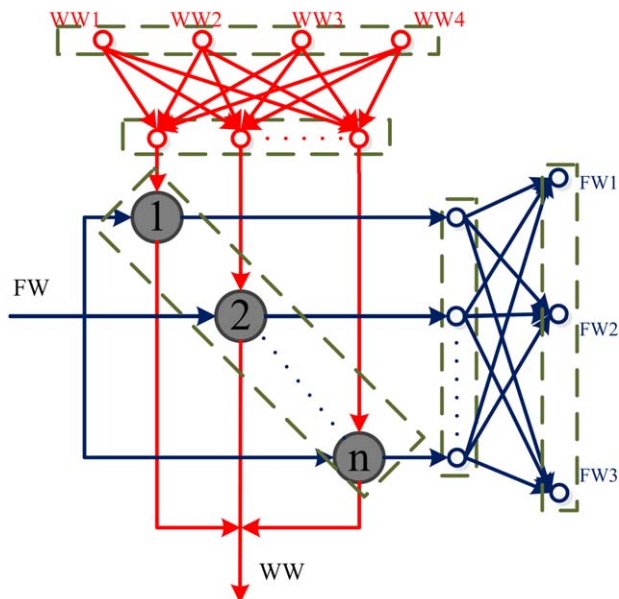


Figure 1. A parallel heat exchange network.

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and heating load of hot and cold streams are exactly identical in the table. For ease of representation, we assume cold streams are listed in an ascending order according to the ending temperature, while hot streams are listed in an ascending order of the starting temperature.

Let's consider using the stream data in Table 1 to construct the parallel HEN structure in Figure 1, where a number of n heat exchangers are organized in parallel. To build the hot and cold streams inside the heat exchangers, both freshwater and wastewater streams have to shape their flow rate and temperature by a series of splitting and mixing scenarios. For such a network structure, one always has to answer two questions: How many heat exchangers are needed and what's the splitting and mixing schemes of freshwater and wastewater streams? Let's start to solve these questions by constructing a new graphical tool, the ΔH -F diagram.

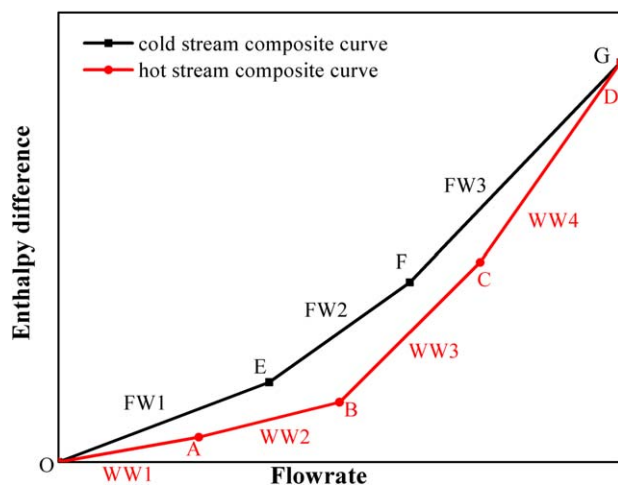


Figure 2. Cold stream composite and hot stream composite curve in ΔH -F diagram.

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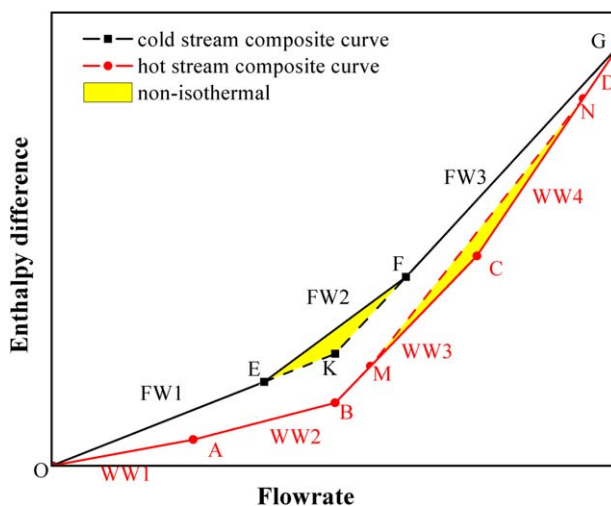


Figure 3. Mixing and splitting of streams in ΔH -F diagram.

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Principle of ΔH -F diagram and its construction

From Figure 1, we can find another feature of the parallel HEN case: the cold end temperature of each heat exchanger is identical as 303 K for hot streams and 293 K for cold streams, respectively. If we use the classical T - ΔH diagram^{23,25} to represent such hot and cold streams, then every stream will start from the same point. This feature made it difficult to figure out the heat exchange matches. Although T - ΔH diagram is powerful in setting energy target, it seems inflexible in the parallel HEN design. Therefore, additional effective graphical tools are needed for the parallel design.

From Figure 1, we can also find three variables to profile both hot and cold streams of each exchanger. They are flow rate, enthalpy change, and temperature difference of hot and cold streams, respectively. These three parameters can be clearly reflected in an ΔH -F diagram, in which flow rate is the abscissa and enthalpy difference is the ordinate. The ΔH -F diagram of Table 1 is shown in Figure 2, in which a hot stream or a cold stream can be presented by line sections. In the figure, if the slope of a hot stream line is identical to that of a cold stream line, the starting temperature of the hot stream line will be 10 K higher than the ending temperature of the cold stream line. In addition, if the length of these two stream lines is also identical, they can form a heat exchange match which holds the 10 K heat transfer temperature approach at both the hot and cold ends. This heat exchange match meets the requirement of the heat exchangers in the parallel network. Therefore, we will synthesis the parallel HEN based on this principle of ΔH -F diagram.

Let's introduce the constructing procedure of Figure 2. In this figure, FW1, FW2, and FW3 are continuously plotted in order of increasing ending temperature with flow rate as the abscissa and enthalpy difference as the ordinate. Note that the order of increasing ending temperature is the same as the slope increasing order, as shown by the corresponding line sections OE, EF, and FG, respectively. We call the combined cold stream line OEFG the cold composite curve. Conversely, WW1, WW2, WW3, and WW4 are presented by line sections OA, AB, BC, and CD, respectively. Similarly, the combined hot stream line OABCD is defined as hot composite curve.

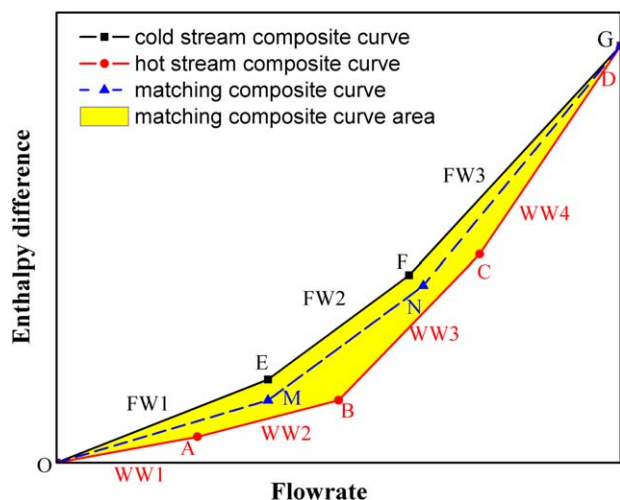


Figure 4. The matching composite curve in ΔH -F diagram.

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Since the stream data in Table 1 reaches overall mass and energy balance, the cold composite curve OEFG and the hot composite curve OABCD can form a closed polygon as shown in the figure: point G overlaps point D.

Mixing and splitting streams in ΔH -F diagram

Because the enthalpy change of the hot and cold streams in Figure 2 are different, they cannot directly match each other in the heat exchangers. They need the splitting and mixing scenarios before or after heat exchange, as shown in Figure 3. Generally, the splitting and mixing procedure are governed by energy and mass balance equations. For example, when a number of n hot/cold streams are mixed to form a new stream, the balance equations can be written as follows

$$F = \sum_{i=1}^n F_i \quad (1)$$

$$F \times T = \sum_{i=1}^n F_i \times T_i \quad (2)$$

where F and T denote the flow rate and temperature of the new stream, while F_i and T_i denote the flow rate and temperature of the i th mixing stream. It is obvious that the above equations can be represented by the closed polygons in the ΔH -F diagram. Therefore, we will design the splitting and mixing scenarios by means of the closed polygons.

The splitting and mixing of cold stream lines is considered first. In Figure 3, cold stream line section EF can be divided into line sections EK and KF, whose starting temperatures are also at 293 K. Since the three line section forms a closed triangle, this division complies with the mass and energy balance. In fact, the cold stream line section can be divided into more than two line sections. No matter how many line sections one cold stream can be divided into, at least one of them has larger slope than the original one and at least one of them has smaller slope than the original one. At this point, if we arrange the divided line sections in slope increasing order, the divided line sections will always be below the original line section. We adopt this slope increasing order to construct the heat exchange matches in the next section. Note that, the flow rate and ending temperature of line sections EK and KF are constrained by the closed triangle.

If the divided line sections have the same slope with the other original line sections, they can be mixed to a single line section. For example, if the slope of line section EK is identical to that of line section OE in Figure 3, then line sections EK and OE can be combined to form new line section OK. The same situation goes for line sections KF and FG, they can also be combined as a new line section KG. As a result, the cold stream composite curve OEFG is transformed to OKG. In other words, three cold streams are transformed to two cold streams.

Figure 3 also shows the splitting and mixing of hot streams. We can see line section BC is split into BM and MC, while CD is split into CN and ND. The ending temperature of the split hot streams BM, MC, CN, and ND are also 303 K. Then, line sections CN and MC are combined to form new line section MN whose ending temperature is also set to 303 K. The closed triangle MCN can be extended to closed polygons when additional mixing streams are added. Since the hot stream line sections are arranged in order of increasing slope, the obtained mixture line section will always be above the original hot stream line sections. The flow rate and starting temperature of the mixture can be directly obtained through the governed equations of the closed polygon.

Heat exchange matches in ΔH -F diagram

Based on the mixing of hot streams and splitting of cold streams, new hot and cold composite curves can be formed. If these two new composites are overlapped section by section, then each section can be used to represent a heat exchange match as mentioned in the previous section. At this point, we call these fully overlapped composite curves the heat exchange matching composite curve (matching composite for

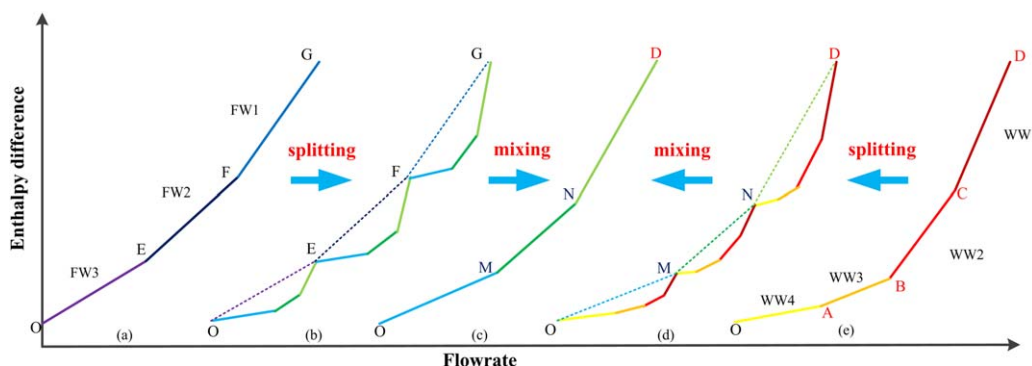


Figure 5. The transformation of the hot/cold stream composite curve to matching composite.

[Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

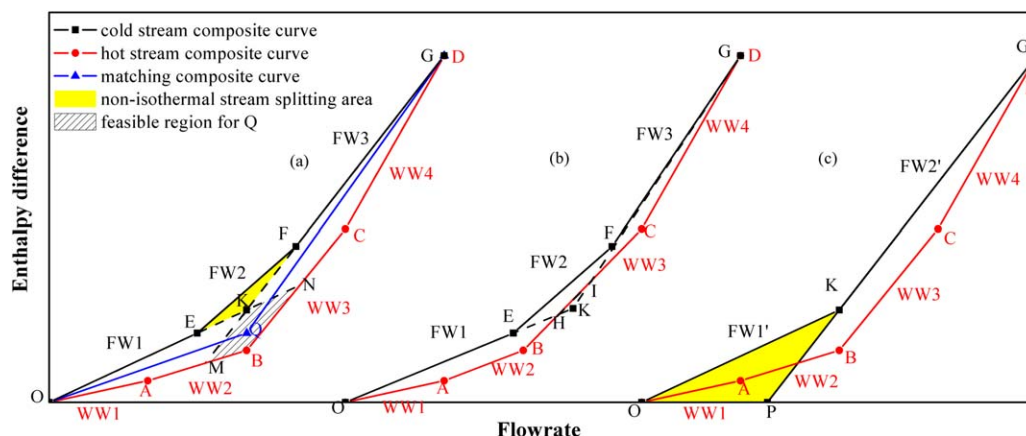


Figure 6. Simplification of HEN.

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short). Since the matching composite is obtained by mixing hot streams or splitting cold streams, it should be located between the original hot and cold stream composite curves. As shown in Figure 4, the dashed line OMNG is one of the possible locations for the matching composite. Once OMNG is determined, both the hot and cold composite curves can be transformed to this composite as shown in Figure 5. For hot composite curve OABCD, each of its line sections are split

into several parts, then these parts are recombined to form the matching composite. This procedure is presented by Figures 5e, d, and c. The transformation of cold composite curve OMNG is in the similar way as illustrated by Figures 5a–c. It should be noted that all the splitting and mixing procedures are governed by the mass and energy balance equations whose solutions are always not unique. In addition, since the parameters for heat exchangers are fixed by the matching composite, different splitting and mixing solutions will not change the total cost. Therefore, any possible solutions are suitable in this case. One can follow several design methods such as the nearest neighborhood algorithm³⁴ and the method proposed by Liao et al.³⁵ It is obvious that the matching composite will approach its limiting position when overlapped with the original hot or cold stream composites.

Note that the above procedure requires cold stream composite being above hot stream composite. Because otherwise one cannot find a heat exchange matching curve that locates between these two composite curves. If part of the cold stream composite is below the hot stream composite, it is requested to adjust the WAN, the utility requirement or the temperature approach to complete the HEN design. These adjustments will be introduced in the case study and the Supporting Information.

Simplification of heat exchange matches

To simplify the heat exchange matches, the number of line sections of the matching composite should be minimized. Suppose a matching composite have a number of n line sections, then the smaller the value of n , the less degree of freedom in constructing the matching composite. In addition, the matching composite is constrained by the cold and hot stream composites. Therefore, there exists a minimum value of n that satisfies the matching requirements. We will start searching this minimum value from a feasible value of n . This value equals the number of cold streams: if the matching composite overlaps with the cold stream composite, it must be a feasible matching composite.

Let's revisit Figure 3. In the figure, the starting composite is the original cold stream composite OEFG, which has a number of three line sections. If we extend the lowest and highest line section OEFG, respectively, they will intersect at point K. the newly formed composite OKG can be designated as matching composite. Consequently, three line sections are decreased to two line sections.

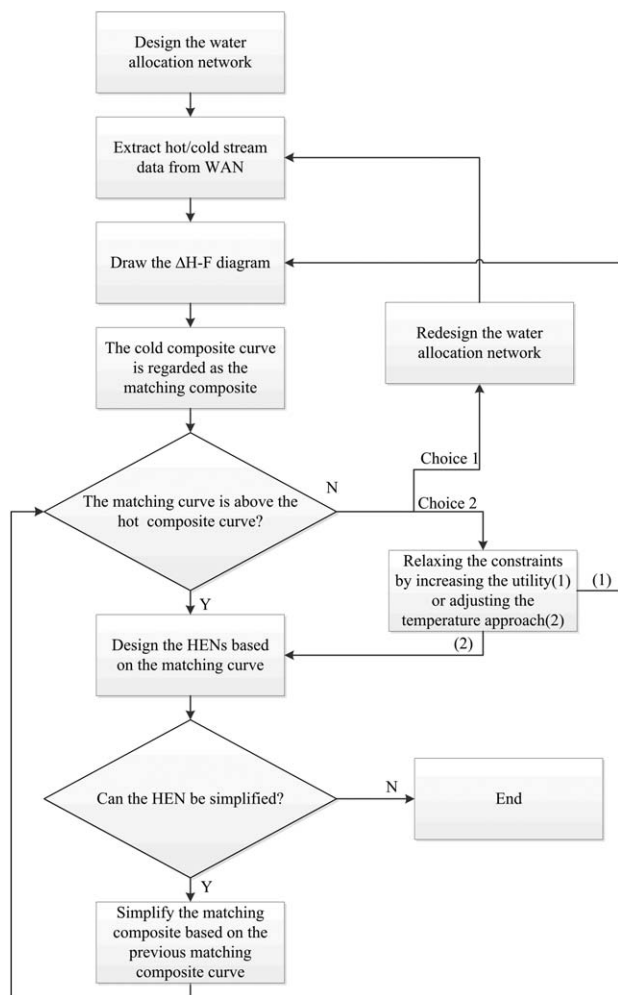


Figure 7. Flow chart of HEN design.

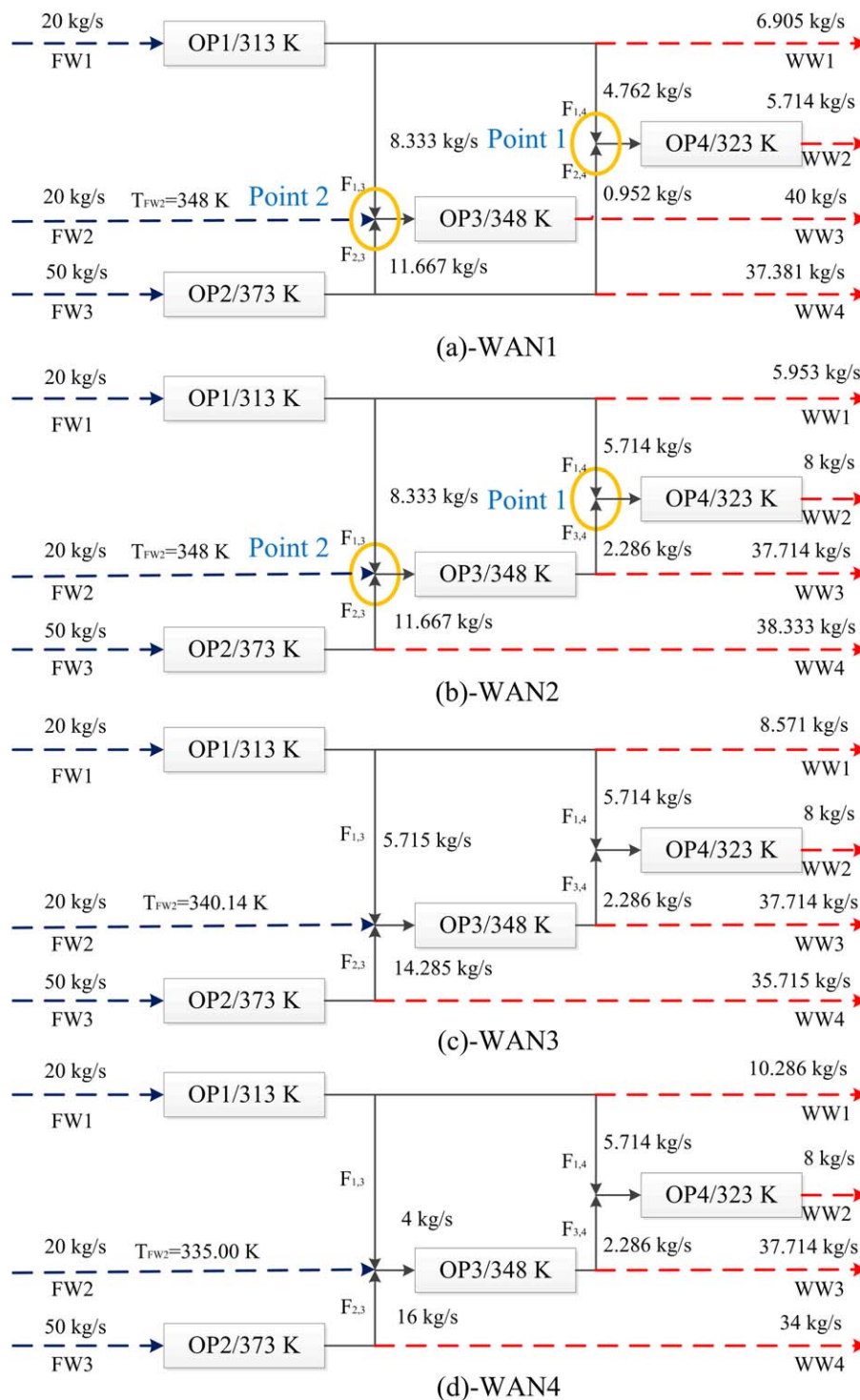


Figure 8. The water allocation networks of example 1.

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Actually, there are many possible matching composites that are composed by two line sections. Without loss of generality, let point Q denote the kink point of such two line sections. In Figure 6a, we extend line section GK to intersect with line section AB at point M, extend line OK to intersect with line section BC at point N. Consequently, a new quadrangle MBNK is formed. If point Q is inside the quadrangle MBNK, then any possible location of matching composite OQG must be between the hot stream composite and cold stream composite.

It should also be noted that point K might be located outside the enclosure of cold and hot composite, as shown in Figure 6b. In this case, additional line sections are needed. As shown in Figure 6b, line sections EK and FK intersect with line section BC at the point H and the point I. Hence, the new matching composite consists of line sections OH, HI and IG. Otherwise, the heat exchange matches represented by matching composite OKG would break the specified minimum temperature approach.

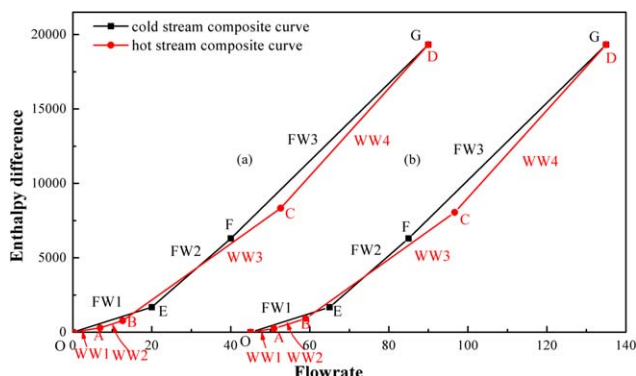


Figure 9. Cold/hot composite curve of the WAN1 and WAN2.

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If the minimum temperature approach is allowed to be lower than 10 K, the number of heat exchange matches can be further simplified to one. The procedure is shown in Figure 6c. In the figure, cold stream line section OK can be split further into line sections OP and PK, whose slope are identical to zero and the slope of KG, respectively. Subsequently, line sections PK and KG are combined together to line section PG, and the matching composite OPG is formed. Because line section OP has no temperature change, only one heat exchange match is needed for composite OPG. It should be noticed that since part of the cold stream composite OPG was below the hot stream composite curve, the minimum temperature approach of 10 K is not satisfied.

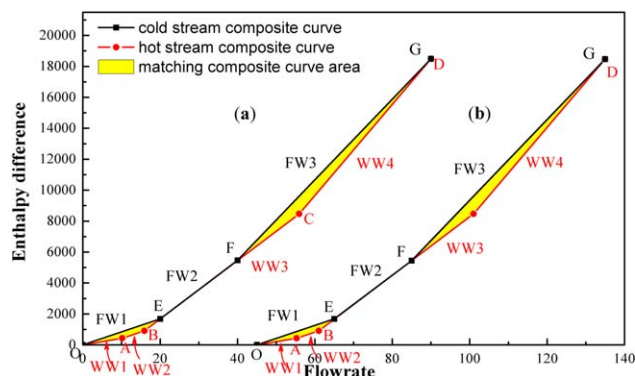


Figure 10. Cold/hot composite curve of the WAN3 (a) ($T_{FW2} = 340.14$ K) and WAN4 (b) ($T_{FW2} = 335$ K).

[Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

Flow chart for HEN design

In summary, there are several steps for the whole design process, as shown in Figure 7. First of all, the water allocation network is obtained by two-dimensional grid diagram.^{23,25} Then, we extract the hot and cold stream data from the WAN. Next, we draw the hot and cold composite curve on the ΔH -F diagram. After that we set the cold composite curve as the initial matching composite. Then, simplify the matching composite by stream mixing and splitting. If the simplified matching composite is entirely above the hot composite curve then we design the HEN based on the matching curve directly. Otherwise, the WAN should be redesigned or the utility/temperature constraints

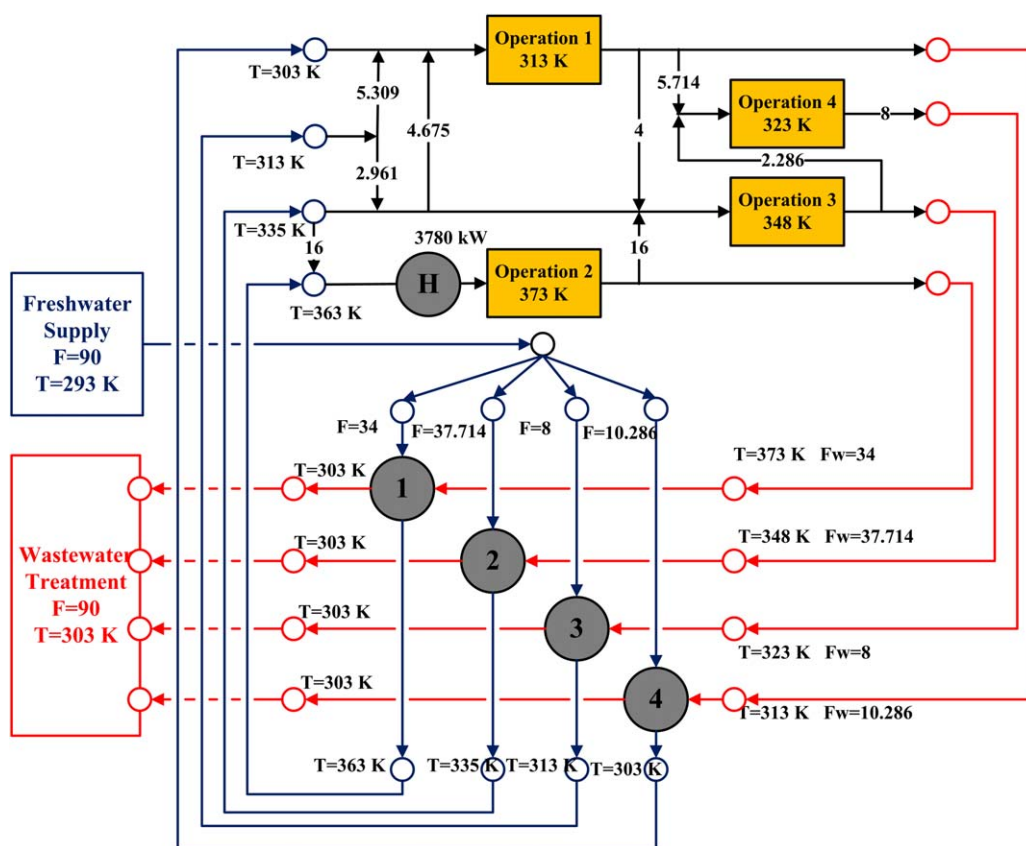


Figure 11. Network structure of example 1-WAN4 ($T_{FW2} = 335$ K) with four heat exchangers.

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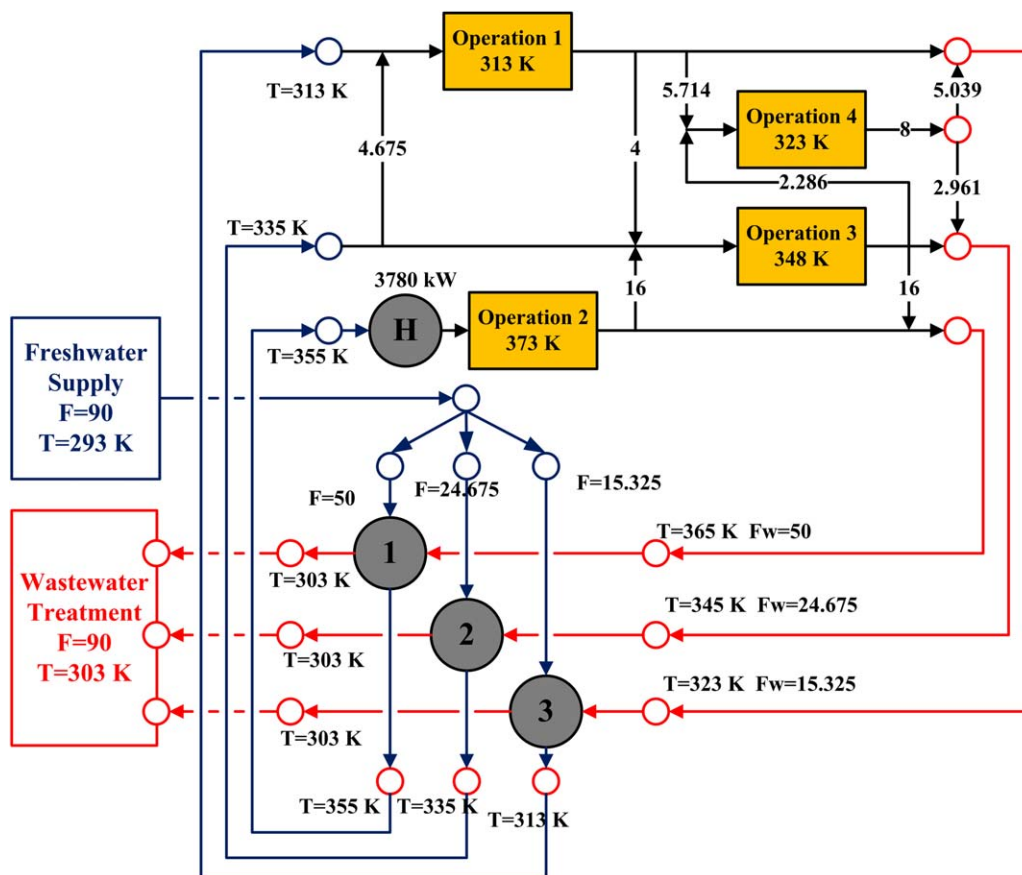


Figure 12. Network structure of example 1-WAN4 ($T_{FW2} = 335$ K) with three heat exchangers.

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should be relaxed. When the number of line sections approaches its minimum value, the whole design process ends.

Case Study

Operating data of these two examples and cost and operating parameters¹⁰ can be found in the Supporting Information.

Example 1

A well-known case study given by Savulescu and Smith^{23,25} is used to illustrate the applicability of the proposed method. This example involves four water using operations. The water

allocation network obtained by two-dimensional grid diagram^{23,25} is presented in Figure 8a, where the red and blue dash lines represents the hot and cold streams, respectively. The data of these hot and cold streams extracted from the water allocation network can be found in the Supporting Information.

From the water allocation network, we can see that the freshwater consumption is 90 kg/s. If the minimum temperature approach of heat transfer between wastewater and freshwater is set to 10 K, hot utility consumption of 3780 kW can be calculated. Hot utility is applied to cold stream FW3 whose ending temperature is the largest. As a result, FW3 is heated

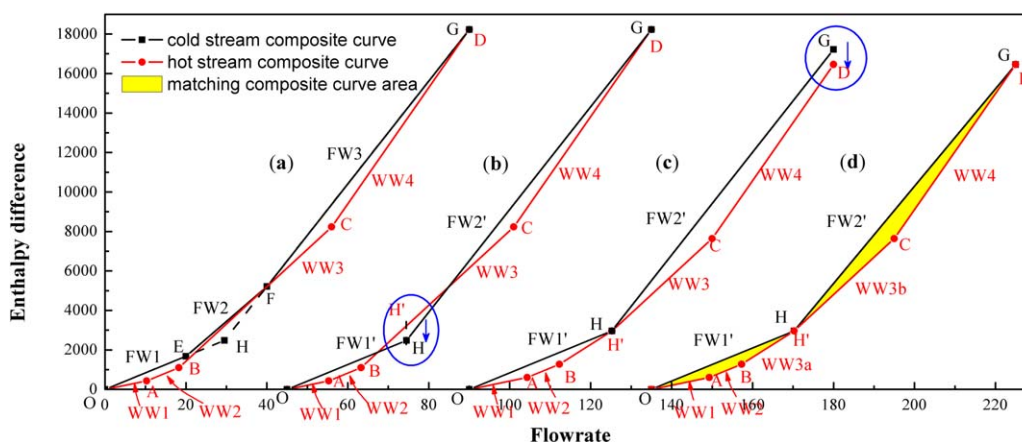


Figure 13. Simplifying HEN procedure 1.

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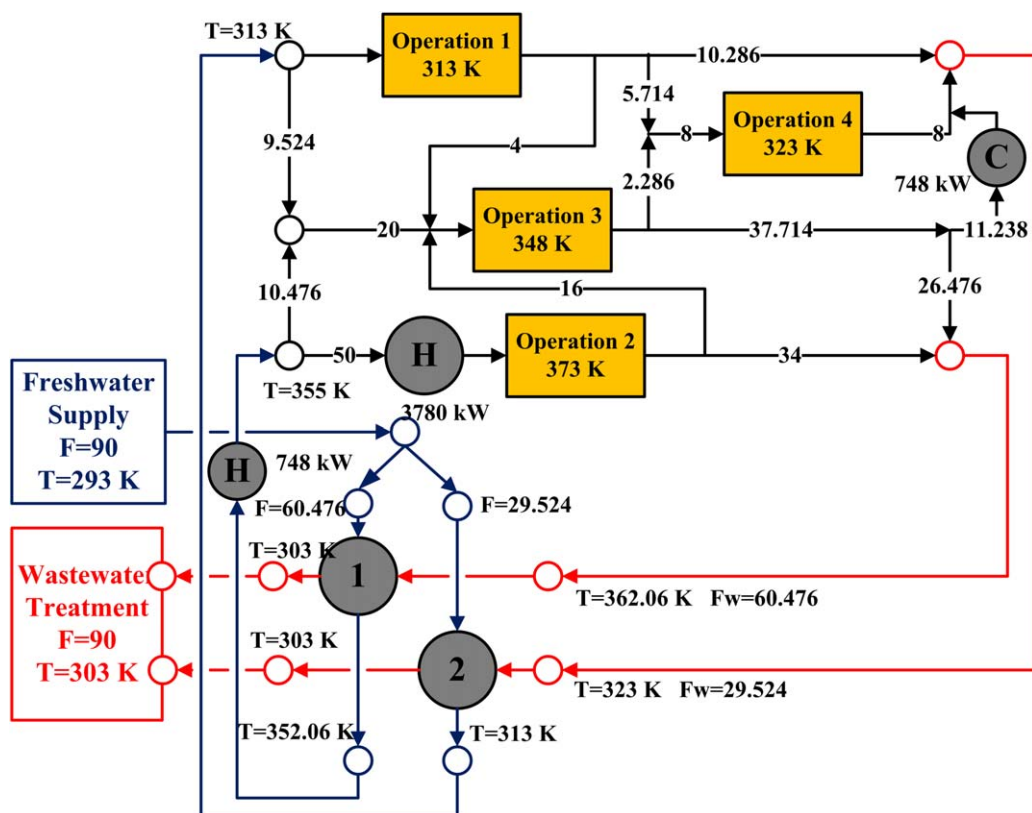


Figure 14. Network structure of example 1 with two heat exchangers obtained by increasing the utility.

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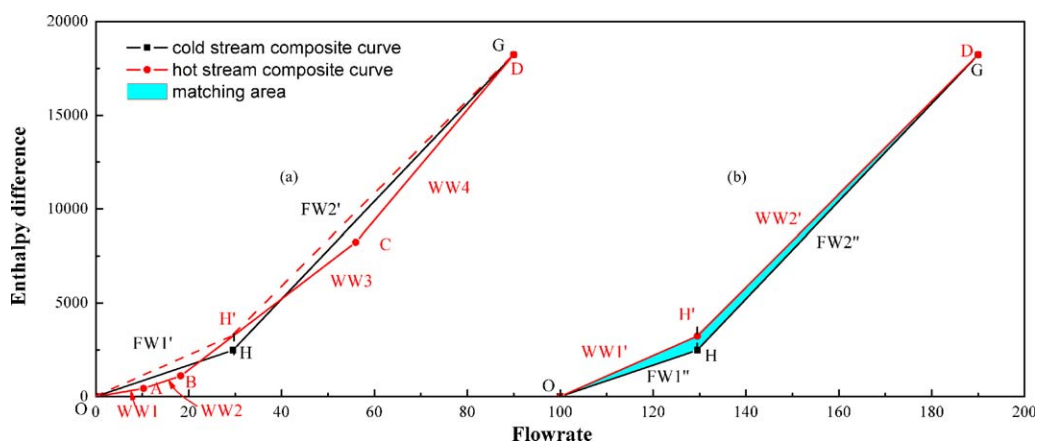


Figure 15. Option 1: keeping the flow rate identical.

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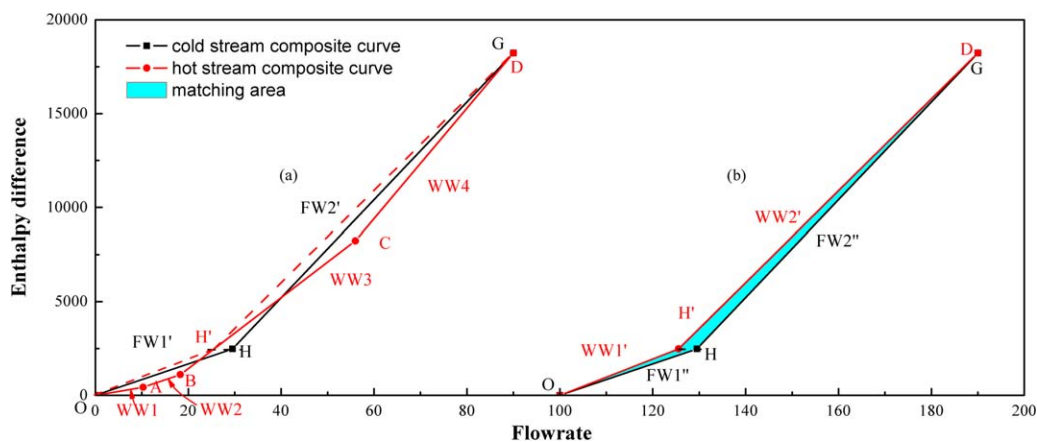


Figure 16. Option 2: keeping the enthalpy difference identical.

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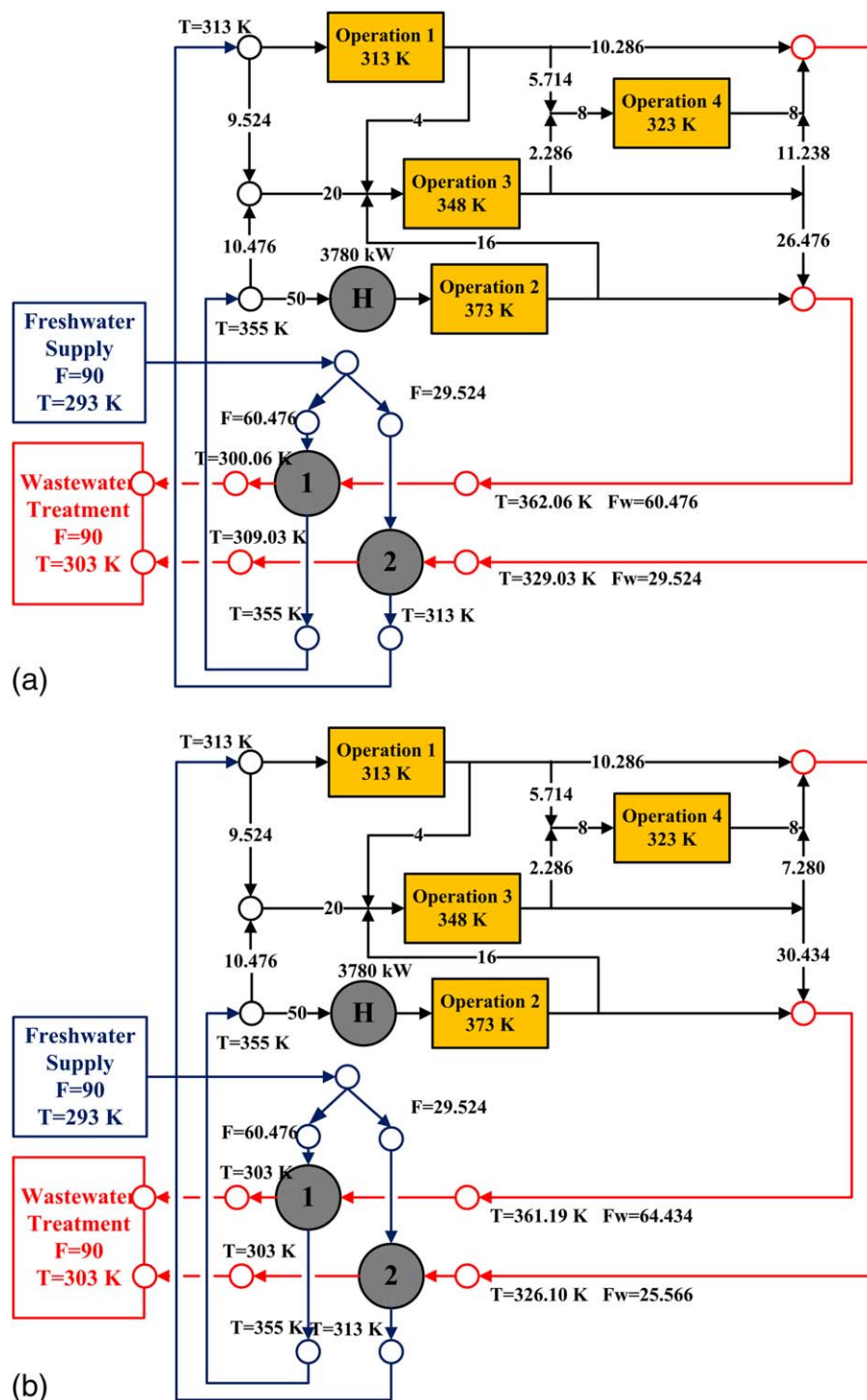


Figure 17. Network structure of example 1 with two heat exchangers obtained by changing the temperature approach.

[Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

up from 293 K to 355 K by wastewater, and is heated up from 355 K to 373 K by hot utility.

Drawing ΔH -F diagram

According to this order, we can draw the hot and cold composite curves in the ΔH -F diagram, as shown in Figure 9a. These two composite curves start from the origin and end in the same point owing to that the sum of the enthalpy differences are equal. In the figure, line sections OA, AB, BC, and CD

represent WW1, WW2, WW3, and WW4, respectively, while line sections OE, EF, and FG represent FW1, FW2, and FW3, respectively.

Adjustment of water allocation networks

Since part of the cold stream composite in Figure 9a is below the hot stream composite curve, it is infeasible to design the HEN with the original utility and temperature approach constraints. As a result, one has to relax these constraints by

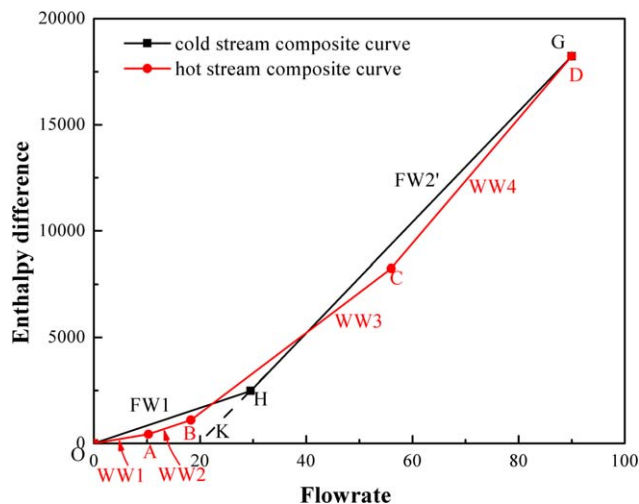


Figure 18. Simplifying HEN procedure 2.

[Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

increasing the utility or decreasing the minimum temperature approach. In fact, there is another way to recover temperature approach: revising the water allocation network. In Figure 9a, we can find one possible way to avoid the crossing of composite curves. That is to enlarge the flow rate of line sections AB (namely WW2) or OA (namely WW1) that are below the crossing area. Let's revisit the water allocation network of Figure 8a to see whether there is such a choice. In Figure 8a, there are two non-isothermal stream mixing points that are related to WW1 or WW2. The first one takes place at the inlet of Operation 4 where reused water $F_{1,4}$ and $F_{2,4}$ are mixed together. The second one occurs at the inlet of Operation 3 where FW2 is mixed with the re-used stream $F_{1,3}$ and $F_{2,3}$. We will adjust the water allocation network from these two points.

1. First point

In Figure 8a, we can increase WW2 by increasing the mixing flow rate of the first point. The current mixing stream $F_{1,4}$ (flow rate from operation 1 to operation 4) and $F_{2,4}$ (flow rate from operation 2 to operation 4) approach their maximum flow rate, because of the energy balance constraint. Therefore, additional

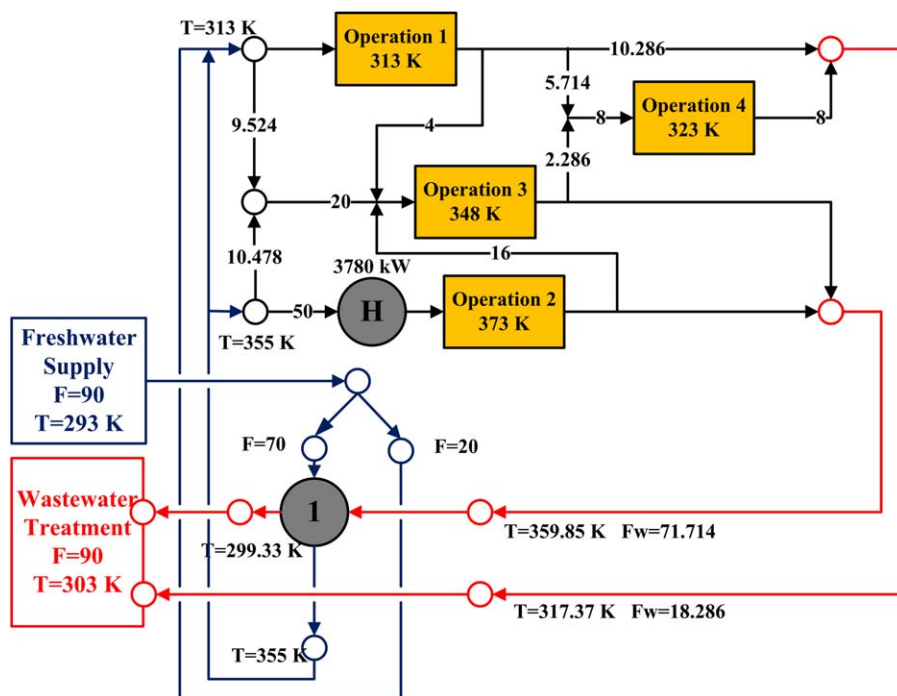


Figure 19. Network structure of example 1 with one heat exchanger.

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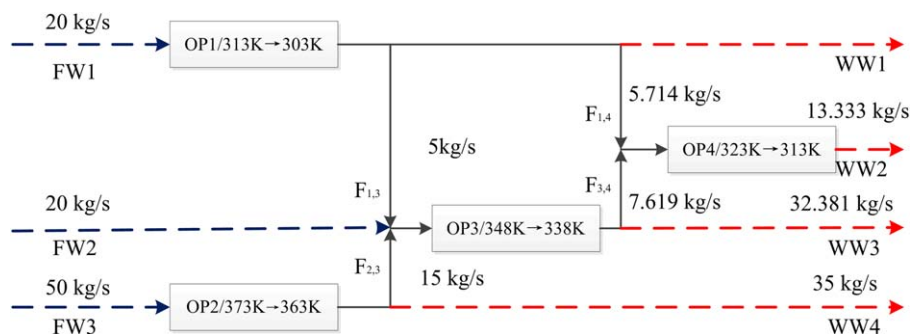


Figure 20. The WAN5 of example 1 considering heat losses.

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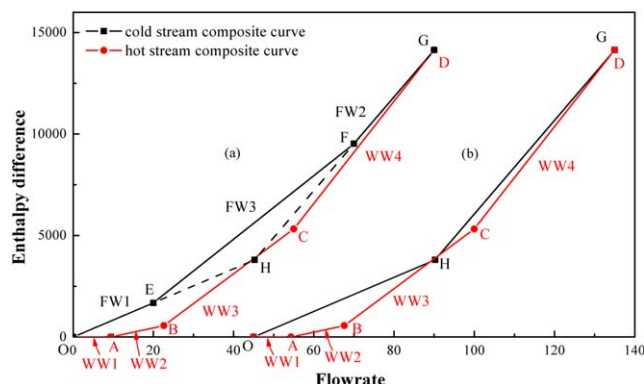


Figure 21. Cold/hot composite curve of the WAN5.

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flow rate from operation 3 is used. One possible solution a new water allocation network, WAN2, is obtained as shown in Figure 8b where the flow rate of WW2 is increased to 8 kg/s. However, we still find the cold composite is above the hot stream composite curve in the ΔH -F diagram, as shown in Figure 9b. Consequently, one has to adjust the second point in WAN2.

2. Second point

We can increase the flow rate of WW1 by decreasing the flow rate of $F_{1,3}$ in the second mixing point of Figure 8b. This will result in the increasing of $F_{2,3}$ and the decreasing of WW4 when the flow rate of FW2 remains unchanged. Such modification will not change the contaminant mass balance, because the concentration of WW1 and WW4 are identical. Conversely, such modifications will change the temperature of FW2 (T_{FW2}) to maintain the energy balance. In fact, T_{FW2} will be decreased as the flow rate of WW1 increases. To figure out the scope of

T_{FW2} in this case, we provide two extreme conditions as shown in Figures 10a, b, where the nonintersected two composite curves touch each other at point E and F, respectively. According to the detailed mass and energy balance calculation, T_{FW2} are 340.14 K and 335 K, respectively, for these two figures. The water allocation networks are shown in Figures 8c, d, respectively. Subsequently, HEN can be designed using these nonintersected composite curves to ensure the temperature approach is 10 K. Let's take Figure 10b as example. If the hot stream composite OABCD and the cold stream composite OEFG are considered as the matching composite, respectively, then the obtained HEN are shown in Figures 11 and 12.

Simplifying the heat exchange networks

We will explore opportunities to further reduce the number of heat exchangers by simplifying matching composites based on the cold stream composite OEFG. As shown in Figure 13a, line sections GF and OE are extended to intersect at point H which means line section EF is split into line sections EH and HF. Then, the cold stream composite OEFG is transformed to the new cold stream composite OHG, including FW1' and FW2' as shown in Figure 13b. Since part of the cold composite is below the hot composite, one has to relax these constraints by increasing the utility or changing the temperature approach. Both of these approaches will be implemented in this example.

1. Increasing the utility

As shown in Figure 13b, draw a vertical line through point H to intersect with line section BC at point H'. Then line section BC (WW3) is divided into two sections BH' (named WW3a) and H'C (named WW3b) whose ending temperatures are also 303 K. The enthalpy difference of line

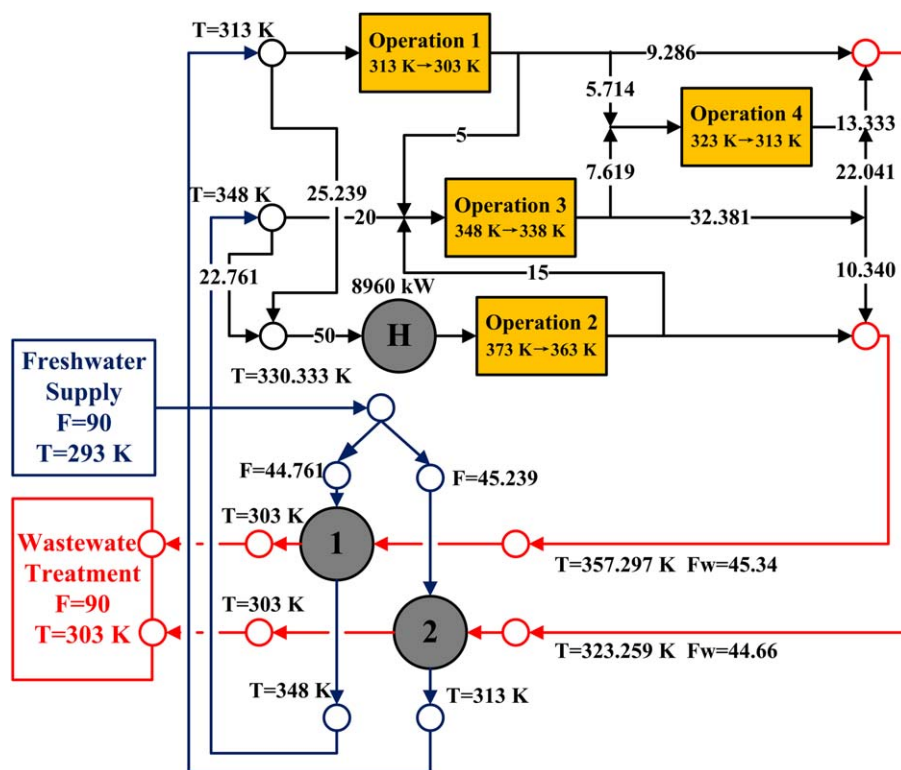


Figure 22. Network structure of example 1 considering heat losses.

[Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

Table 2. The Result for Freshwater Consumption, the Utility, the Total Heat Flow Exchanged in the Heat Exchangers, and the Total Area of Heat Exchangers-Example 1

Method proposed by the authors	FW (kg/s)	HU (kW)	CU (kW)	Equipment Requirements(/)	$Q_{total}(10^4\text{kW})$	$A_{total}(10^3\text{m}^2)$
Ref. 1/Savulescu et al. ²³	90	4265	485	3 heat exchanger, 1 heater, 1 cooler	2.359	4.548
Ref. 2/Liao et al. ¹⁵	90	3780	0	4 heat exchanger, 1 heater	2.084	4.472
Ref. 3/Leewongtanawit et al. ²⁵	90	3780	0	3 heat exchanger, 1 heater	2.226	3.775
Ref. 4/Ahmetović et al. ¹⁴	90	3780	0	2 heat exchanger, 1 heater	2.234	3.960
Ref. 5/Ahmetović et al. ^{14*}	90	3780	0	1 heat exchanger, 1 heater	2.369	6.072
Ref. 6/Savulescu et al. ²³	90	8640	0	3 heat exchanger, 1 heater	2.310	3.514
This article						
Case 1/Figure 11	90	3780	0	4 heat exchanger, 1 heater	2.201	3.915
Case 2/Figure 12	90	3780	0	3 heat exchanger, 1 heater	2.201	3.907
Case 3/Figure 14	90	4528	748	2 heat exchanger, 2 heater, 1 cooler	2.276	3.832
Case 4/Figure 17a	90	3780	0	2 heat exchanger, 1 heater	2.201	5.040
Case 5/Figure 17b	90	3780	0	2 heat exchanger, 1 heater	2.201	4.667
Case 6/Figure 19	90	3780	0	1 heat exchanger, 1 heater	2.201	6.830
Case 7/Figure 22	90	8960	0	2 heat exchanger, 1 heater	2.310	3.375

Note that all flow rates in Figure 7 of Ahmetović et al.¹⁴ multiply 3.6.

Table 3. The Results for the Freshwater Cost, the Utilities Cost, the Investment Cost, and the Total Annual Cost-Example 1

Method proposed by the authors	FWC($10^5\$/\text{yr}$)	HUC($10^6\$/\text{yr}$)	CUC($10^4\$/\text{yr}$)	IC($10^5\$/\text{yr}$)	TAC($10^6\$/\text{yr}$)
Ref. 1/Savulescu et al. ²³	9.720	1.608	9.167	3.692	3.041
Ref. 2/Liao et al. ¹⁵	9.720	1.425	0	3.401	2.737
Ref. 3/Leewongtanawit et al. ²⁵	9.720	1.425	0	3.103	2.707
Ref. 4/Ahmetović et al. ¹⁴	9.720	1.425	0	2.559	2.653
Ref. 5/Ahmetović et al. ¹⁴	9.720	1.425	0	2.710	2.668
Ref. 6/Savulescu et al. ²³	9.720	3.352	0	3.069	4.631
This article					
Case 1/Figure 11	9.720	1.425	0	3.230	2.720
Case 2/Figure 12	9.720	1.425	0	2.961	2.693
Case 3/Figure 14	9.720	1.707	1.414	2.902	3.111
Case 4/Figure 17a	9.720	1.425	0	2.817	2.679
Case 5/Figure 17b	9.720	1.425	0	2.773	2.674
Case 6/Figure 19	9.720	1.425	0	2.845	2.682
Case 7/Figure 22	9.720	3.378	0	2.562	4.606

section HH' reads 748 kW. To eliminate the crossover areas, the enthalpy change of BH' decreases the same amount. In other words, BH' is cooled to 331.90 K before heat exchanging with freshwater streams. This is illustrated by Figure 13c, where H' is moved to the position of point H. From the Figure 13c, we can see that additional heating loads are required by the cold composite curve. Therefore, an equivalent hot utility

is added to cold stream FW3 whose ending temperature is the largest. As a result, FW2' is heated up to 352.06 K by wastewater, and then heated up to 373 K by hot utility. With the modified line of FW2', the composite curves are shown in Figure 13d in which the hot composite curve OABH'CD is completely below the cold composite curve OHG.

When the cold stream composite OHG is recognized as the matching composite, the hot stream composite OABH'CD can be transformed to satisfy the two specified matches GH and HO by mixing. The mixing schemes are clearly shown in Figure 13d as the two closed polygons: WW4 and WW3b are mixed to feed match GH, while WW3a, WW2, and WW1 are mixed to feed match HO. Consequently, the HEN with two heat exchangers is obtained, as shown in Figure 14.

2. Changing the temperature approach

Option 1: keeping the flow rate identical

The cold stream composite OHG in Figure 15a is considered as the matching composite. Then, draw a vertical line through point H to intersect with line section BC at point H', connect point H' and point D, and connect point O and point H'. Subsequently, a new hot stream composite OH'G(D) can be obtained, as shown in Figure 15b. There are two heat exchange matches, OH'/OH and H'D/HG. It should be noted that the enthalpy difference is not balanced in these two matches OH'/OH and H'D/HG. To rebalance the enthalpy differences, we modify the ending temperature of hot streams. For match

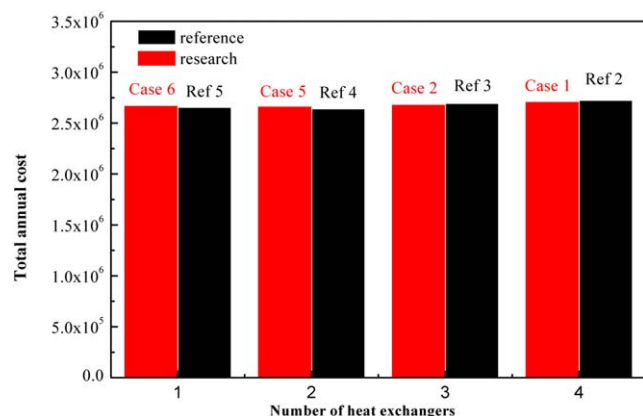


Figure 23. Total annual cost comparison of the cases.

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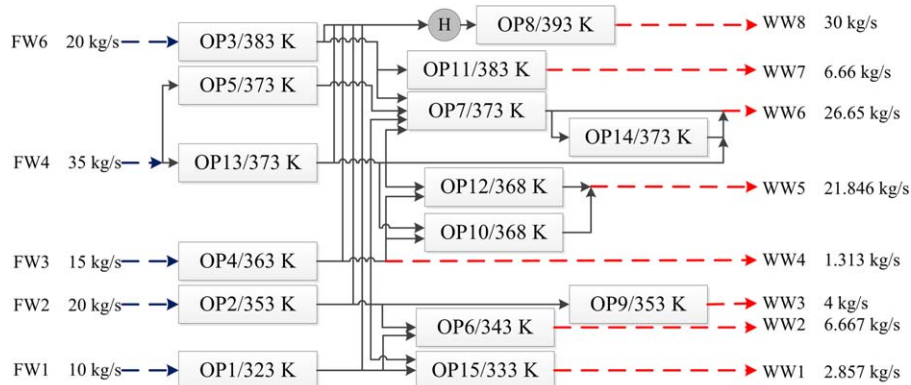


Figure 24. The WAN of example 2 obtained by two-dimensional grid diagram.

[Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

OH'/OH, the temperature difference of cold stream OH is 20 K, while the starting temperature of hot stream OH' is 329.03 K. Consequently, the ending temperature of stream OH' is shifted from 303 K to 309.03 K to keep the temperature difference of 20 K. Similarly, the ending temperature of H'D is modified from 303 K to 300.06 K to keep the stream temperature difference of 62 K (cold stream EF is heated from 293 K to 355 K). The temperature approach for matches OH'/OH and H'D/HG are 16.03 K and 7.06 K, respectively. The enthalpy change of both streams OH' and H'D are modified. However, the overall energy balance remains unchanged: the mixing temperature of the hot stream OH' and the hot stream H'D is still 303 K. As a result, the HEN with two heat exchangers is obtained, as shown in Figure 17a.

Option 2: keeping the enthalpy difference the same

The cold stream composite OHG is also recognized as the matching composite in Case 2. Compared to Case 1, we will reshape the hot stream composite in another way. In Figure 16a, a horizontal line is drawn through point H to intersect with line section BC at point H'. Points O and H', and points H' and D are connected. Subsequently, a new hot stream composite OH'D(G) can be obtained, as shown in the Figure 16b. In the figure, we find two heat exchange matches, OH'/OH and H'D/HG. All of them are balanced in enthalpy change. But the flow rate is not balanced. Since the enthalpy difference is identical, because the flow rate is not specified identical in this case, we can directly check the temperature approach of each heat exchange matches. The temperature approach of heat

exchanger for OH'/OH is 13.10 K in the hot end and 10 K in the cold end. The temperature approach of heat exchanger for H'D/HG is 6.19 K in the hot end and 10 K in the cold end. As a result, the HEN with two heat exchangers is obtained, as shown in Figure 17b.

In fact, the HEN can be simplified further to only one heat exchanger, as shown in Figure 18. Let's extend line section GH to intersect with the abscissa at point K which means that line section OH is split to line sections OK and KH. As a result, the cold stream composite OHG is transformed to OKG. The cold stream composite OKD is considered as matching composite. There is only one heat exchanger needed for the cold stream line section KG since line section OK represents freshwater. Hot streams BC and CD are mixed to match with the cold stream KG. The HEN with one heat exchanger is obtained, as shown in Figure 19.

Heat losses from the operations

In the previous part, all water-using operations have been considered as isothermal. However, there might be heat losses within the operations in practice. Hence, in this section, we will introduce the HEN design considering water-using operations with heat losses. These operations are characterized by a temperature difference between inlet stream and outlet stream. We assume that the temperature of the outlet stream is 10 K lower than the one of the inlet stream for all operations. Then, two-dimensional grid diagram^{23,25} are adopted to design the water allocation network, presented in Figure 20. After that, stream data are extracted from the water network. Hot utility consumption of 8960 kW can be calculated according to the freshwater consumption and the heat losses of all operations. Based on the methodology of ΔH -F diagram, hot utility is applied to the cold stream FW3 whose ending temperature is the largest. Then, the hot and cold composite curves in the ΔH -F diagram are obtained, as shown in Figure 21a. In the figure, line sections OA, AB, BC, and CD represent WW1, WW2, WW3, and WW4, respectively, while line sections OE, EF, and FG represent FW1, FW2, and FW3, respectively. It is apparent that the cold stream composite curve OEFG is above the hot stream curve OABCD, one can obtain HEN with three heat exchangers that satisfy the specified temperature approach constraint. To further reduce the number of heat exchangers by simplifying matching composites based on the cold stream composite OEFG, line sections OE and GF are extended to intersect at the point H. The cold composite curve

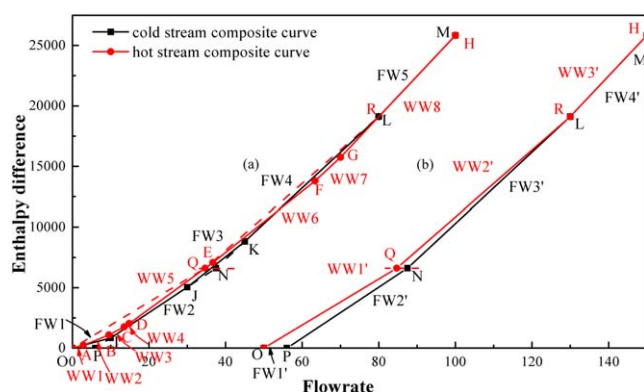


Figure 25. Cold/hot composite curve of example 2.

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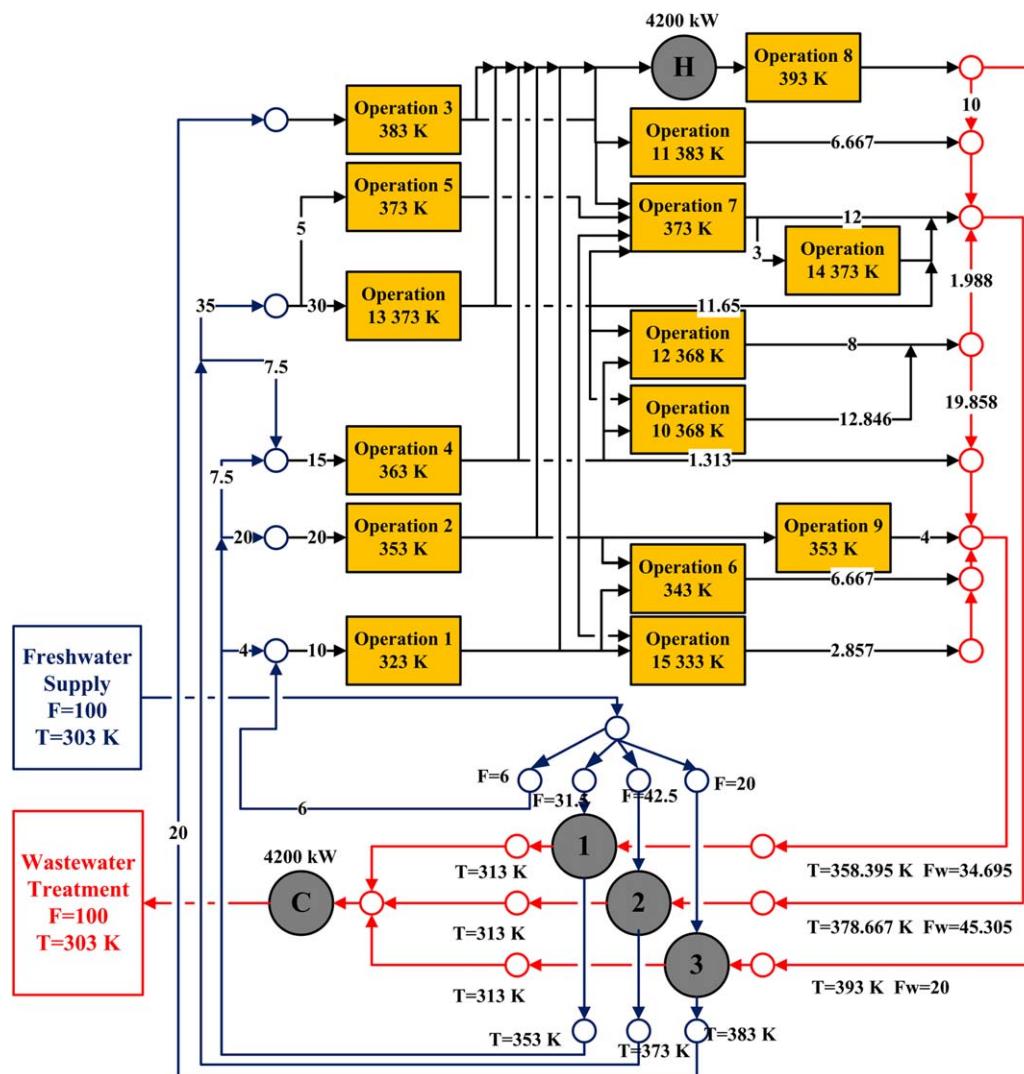


Figure 26. Network structure of example 2 with three heat exchangers.

[Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

OEFG is transformed to OHG, as shown in Figure 21b. It is easily found that part of the cold composite is below the hot composite, one has to relax these constraints by increasing the utility or changing the temperature approach. In this example, changing the temperature approach is adopted, the HEN with two heat exchangers is obtained, as shown in Figure 22.

This example has been solved by several authors using different approaches based on conceptual design or mathematical programming. To compare the result of this work with the literature results, all cost coefficients are unified by the data presented in the Supporting Information. Compared to those networks in the literature, the networks obtained in this article are unique in parallel HEN structure. The network structures obtained in the previous section are named Case 1 (Figure 11), Case 2 (Figure 12), Case 3 (Figure 14), Case 4 (Figure 17a),

Case 5 (Figure 17b), Case 6 (Figure 19), and Case 7 (Figure 22), respectively, according to the order of appearance.

The freshwater and utility consumptions, as well as equipment scale requirements are summarized in Table 2, while the operating and capital cost comparisons are listed in Table 3.

Among these compared networks, freshwater consumptions are all minimized to 90 kg/s. In addition, the hot and cold utility consumptions are also minimized to 3780 kW and 0 kW, respectively, except for the network of Savulescu et al.²³ and Case 3. Although the total heat exchanger area of Case 3 is small, its total annual cost is still large owing to the larger utility consumption. Note that, when heat losses is considered, the Case 7 of this study has lower TAC than the case (Ref. 6) in Savulescu et al.²³ owing to using less heat exchangers.

Table 4. The Results for the Freshwater Cost, the Utilities Cost, the Investment Cost, and the Total Annual Cost-Example 2

Method Proposed by the Authors	FW (kg/s)	HU (kW)	CU (kW)	Equipment Requirements (/)	FWC (10 ⁶ \$/yr)	HUC (10 ⁶ \$/yr)	CUC (10 ⁵ \$/yr)	IC (10 ⁵ \$/yr)	TAC (10 ⁶ \$/yr)
Case 3/Liu et al. ²¹	100	4200	4200	6 heat exchanger, 2 heater, 1 cooler	1.080	1.630	7.938	5.236	4.027
This article case/figure 29	100	4200	4200	3 heat exchanger, 1 heater, 1 cooler	1.080	1.630	7.938	4.643	3.968

The total annual cost (TAC) comparison can be summarized in Figure 23, where the red columns represent the cases in this research while the black columns represent the results in the references. The comparison is organized by the number of heat exchangers. When the results include a number of 3 or 4 heat exchangers, this work performs better than the previous works.^{10,13} When the results include a number of only 1 or 2 heat exchangers, the TAC results in Ahmetović et al.¹⁴ are 0.52% to 0.79% better than this work. From Figure 23, we can see that the TAC differences between cases are small. However, the network structures are totally different between the cases. The results obtained by this work are HEN in parallel, while the others are HEN in series. Consequently, the results of this work will be more reliable when considering accidental shutdown of heat exchangers.

Example 2

The third example is a large system consist of 15 operations from Luo et al.,³⁰ which is the largest example in the existing literature. Note that the temperature of freshwater and wastewater both are 303 K. Hence, one has to divide this problem into two parts. In the first part, the temperature of wastewater is considered as 313 K. In the second part, one cooler must be used to cool down all wastewater to 303 K. The proposed method can be applied for the first part. Two-dimensional grid diagram^{23,25} is adopted to design the water allocation network, as shown in Figure 24. The data of these hot and cold streams extracted from the Figure 24 can be found in the Supporting Information. The freshwater consumption is 100 kg/s. Hence, the hot utility of 4200 kW is calculated. Note that, since operating temperature of operation 8 is highest, hot utility is applied to the stream flowing into the operation 8. Next, the hot composite curve OABCDEFGH and the cold composite curve OIJKLM are drawn on the ΔH -F diagram, as shown in Figure 25a. Then, the matching curve is constructed based on the cold composite curve. Extend line sections LK and IJ to intersect at point N, extend line section JI to intersect with the abscissa at point P. As a result, the original cold composite curve is transformed to OPNLM. Since the slopes of line sections GH and LM are the same, line section GH is split into GR and RH to make RH equal to LM. Draw a horizontal line through point N to intersect with line section CF at point Q, connect point O and point Q, and connect point Q and point R. The original hot composite curve is transformed to OQRH. Consequently, there are three heat exchange matches: OQ/PN, QR/NL, and RH/LM, as shown in Figure 25b. The HEN with three heat exchangers is obtained, as shown in Figure 26.

The cost and operating parameters are applied to this case, the total annual cost of $\$3.968 \times 10^6/\text{yr}$ is calculated. This example is also solved by Liu et al.²¹ To compare the result of this article with their result, same cost coefficients are used to recalculate the total annual cost. Table 4 shows the summarized cost comparisons. Note that the number of heat exchangers as well as the TAC obtained in this article is smaller than those obtained by Liu et al.²¹

Conclusion

This article addressed the HEN design of WAHEN by a novel graphical approach. The proposed method features two important properties. First, the mixing and splitting of streams in ΔH -F diagram is visual and accessible. The mixing and splitting procedures are guided by the specified matching composite curves to form desired HEN structures. The HEN structure can

also be simplified by modifying the matching composites. Second, the HEN obtained by the proposed method are parallel structure, while most of HEN obtained by conceptual design in the previous literature are series structure. Parallel structure is simpler especially when the number of streams is large. Furthermore, parallel structure is feasible for distributed effluent-treatment systems, and it might be more reliable when freshwater or wastewater flow rate fluctuates.

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Notation

Parameter

A_{total}	= total area of heat exchangers
C_{IN}	= limiting inlet concentration, ppm
C_{OUT}	= limiting outlet concentration, ppm
CU	= cold utility consumption
CUC	= cold utility cost
F	= flow rate
FW	= freshwater consumption
FWC	= freshwater cost
HU	= hot utility consumption
HUC	= hot utility cost
IC	= investment cost
Q_{total}	= total heat flow exchanged in the heat exchangers
TAC	= total annual cost
$T_{\text{OP,IN}}$	= inlet water temperature of operation, K
$T_{\text{OP,OUT}}$	= outlet water temperature of operation, K
ΔH	= enthalpy difference

Subscript

FW	= freshwater
WW	= wastewater

Literature Cited

1. Bagajewicz M. A review of recent design procedures for water networks in refineries and process plants. *Comput Chem Eng.* 2000; 24(9–10):2093–2113.
2. Foo DCY. State-of-the-art review of pinch analysis techniques for water network synthesis. *Ind Eng Chem Res.* 2009;48(11):5125–5159.
3. Jezowski J. Review of water network design methods with literature annotations. *Ind Eng Chem Res.* 2010;49(10):4475–4516.
4. Furman KC, Sahinidis NV. A critical review and annotated bibliography for heat exchanger network synthesis in the 20th century. *Ind Eng Chem Res.* 2002;41(10):2335–2370.
5. Bagajewicz M, Rodera H, Savelski M. Energy efficient water utilization systems in process plants. *Comput Chem Eng.* 2002;26(1):59–79.
6. Leewongtanawit B, Kim J-K. Synthesis and optimisation of heat-integrated multiple-contaminant water systems. *Chem Eng Process.* 2008;47(4):670–694.
7. Alva-Argáez A, Kokossis A, Smith R. Wastewater minimisation of industrial systems using an integrated approach. *Comput Chem Eng.* 1998;22:S741–S744.
8. Papoulias SA, Grossmann IE. A structural optimization approach in process synthesis—II: heat recovery networks. *Comput Chem Eng.* 1983;7(6):707–721.
9. Floudas C, Ciric A. Strategies for overcoming uncertainties in heat exchanger network synthesis. *Comput Chem Eng.* 1989;13(10): 1133–1152.

10. Dong H-G, Lin C-Y, Chang C-T. Simultaneous optimization approach for integrated water-allocation and heat-exchange networks. *Chem Eng Sci.* 2008;63(14):3664–3678.
11. Rui-Jie Z, Li-Juan L, Hong-Guang D, Ignacio IE. Synthesis of inter-plant water-allocation and heat-exchange networks. Part 1: fixed flow rate processes. *Ind Eng Chem Res.* 2012;51(11):4299–4312.
12. Rui-jie Z, Li-juan L, Hong-Guang D, Ignacio IE. Synthesis of inter-plant water allocation and heat exchange networks. Part 2: integrations between fixed flow rate and fixed contaminant-load processes. *Ind Eng Chem Res.* 2012;51:14793–14805.
13. Bogataj M, Bagajewicz MJ. Synthesis of non-isothermal heat integrated water networks in chemical processes. *Comput Chem Eng.* 2008;32(12):3130–3142.
14. Ahmetović E, Kravanja Z. Simultaneous synthesis of process water and heat exchanger networks. *Energy.* 2013;57:236–250.
15. Liao ZW, Rong G, Wang JD, Yang YR. Systematic optimization of heat-integrated water allocation networks. *Ind Eng Chem Res.* 2011; 50(11):6713–6727.
16. Yee TF, Grossmann IE. Simultaneous optimization model for heat integration—II: heat exchanger network synthesis. *Comput Chem Eng.* 1990;14(10):1165–1184.
17. Ahmetović E, Kravanja Z. Simultaneous optimization of heat-integrated water networks involving process-to-process streams for heat integration. *Appl Thermal Eng.* 2014;62(1):302–317.
18. Ahmetović E, Ibrić N, Kravanja Z. Optimal design for heat-integrated water-using and wastewater treatment networks. *Appl Energy.* 2014;135:791–808.
19. Ibrić N, Ahmetović E, Kravanja Z. Two-step mathematical programming synthesis of pinched and threshold heat-integrated water networks. *J Clean Prod.* 2014;77:116–139.
20. Ibrić N, Ahmetović E, Kravanja Z. Simultaneous optimization of water and energy within integrated water networks. *Appl Thermal Eng.* 2014;70:1097–1122.
21. Liu ZM, Luo YQ, Yuan XG. Simultaneous integration of water and energy in heat-integrated water allocation networks. *AIChE J.* 2015; 61(7): 2202–2214.
22. Ghazouani S, Zoughaib A, Pelloux-Prayer S. Simultaneous heat integrated resource allocation network targeting for total annual cost considering non-isothermal mixing. *Chem Eng Sci.* 2015;134:385–398.
23. Savulescu L, Kim J-K, Smith R. Studies on simultaneous energy and water minimisation—Part II: systems with maximum re-use of water. *Chem Eng Sci.* 2005;60(12):3291–3308.
24. Savulescu L, Kim J-K, Smith R. Studies on simultaneous energy and water minimisation—Part I: systems with no water re-use. *Chem Eng Sci.* 2005;60(12):3279–3290.
25. Leewongtanawit B, Kim J-K. Improving energy recovery for water minimisation. *Energy.* 2009;34(7):880–893.
26. Sahu GC, Bandyopadhyay S. Energy optimization in heat integrated water allocation networks. *Chem Eng Sci.* 2012;69(1):352–364.
27. Manan ZA, Tea SY, Alwi SRW. A new technique for simultaneous water and energy minimisation in process plant. *Chem Eng Res Des.* 2009;87(11):1509–1519.
28. Wan Alwi SR, Ismail A, Manan ZA, Handani ZB. A new graphical approach for simultaneous mass and energy minimisation. *Appl Thermal Eng.* 2011;31(6–7):1021–1030.
29. Yiqing L, Tingbi M, Sucai L, Xigang Y. Studies on the effect of non-isothermal mixing on water-using network's energy performance. *Comput Chem Eng.* 2012;36:140–148.
30. Luo Y, Liu Z, Luo S, Yuan X. Thermodynamic analysis of non-isothermal mixing's influence on the energy target of water-using networks. *Comput Chem Eng.* 2014;61:1–8.
31. Feng X, Li Y, Shen R. A new approach to design energy efficient water allocation networks. *Appl Thermal Eng.* 2009;29(11–12):2302–2307.
32. Hou Y, Wang J, Chen Z, Li X, Zhang J. Simultaneous integration of water and energy on conceptual methodology for both single- and multi-contaminant problems. *Chem Eng Sci.* 2014;117:436–444.
33. Liu ZY, Yang Y, Wan LZ, Wang X, Hou KH. A heuristic design procedure for water-using networks with multiple contaminants. *AIChE J.* 2009;55(2):374–382.
34. Prakash R, Shenoy UV. Targeting and design of water networks for fixed flowrate and fixed contaminant load operations. *Chem Eng Sci.* 2005;60(1):255–268.
35. Liao ZW, Lou JY, Wang JD, Jiang BB, Yang YR. Mixing potential: a new concept for optimal design of hydrogen and water networks with higher disturbance resistance. *AIChE J.* 2014;60(11):3762–3772.

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